The activation of the n^{th} neuron (or, more accurately in this case, layer) is the product of the $(n-1)^{\text{th}}$ weight matrix $\mathbf{W}$ and prior activation $\vec{a}$. Let $\vec{a}_n$ be the output layer, and let $\vec{a}_1$ be the input layer.

Let:

$$\underbrace{\mathbf{W}_{n-1}}_{q\times p} : \underbrace{\vec{a}_{n-1}}_{p\times 1} \mapsto \underbrace{\vec{a}_n}_{q\times 1}.$$

Let the cost function be defined as:

$$\begin{aligned} C &= (\vec{a}_n - \vec{e})^2 \\ &= (\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e})^2 \\ &= (\sigma(\mathbf{W}_{n-1}\sigma(\mathbf{W}_{n-2}\vec{a}_{n-2})) - \vec{e})^2 \\ &\vdots \end{aligned}$$

Thus:

$$\begin{aligned} \underbrace{\frac{\partial C}{\partial \mathbf{W}_{n-1}}}_{q\times p} &= \underbrace{2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e})}_{q\times 1} \odot \underbrace{\sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1})}_{q\times 1} \underbrace{\vec{a}_{n-1}^{\mathrm{T}}}_{1\times p} \\ &= \delta_{n-1}\vec{a}_{n-1}^{\mathrm{T}}. \end{aligned}$$

Consider $n-2$:

$$\underbrace{\mathbf{W}_{n-2}}_{p\times r} : \underbrace{\vec{a}_{n-2}}_{r\times 1} \mapsto \underbrace{\vec{a}_{n-1}}_{p\times 1}.$$

$$\begin{aligned} \frac{\partial C}{\partial \mathbf{W}_{n-2}} &= 2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e}) \odot \sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1}) \frac{\partial \left[\mathbf{W}_{n-1}\vec{a}_{n-1}\right]}{\partial \mathbf{W}_{n-2}} \\ &= 2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e}) \odot \sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1}) \frac{\partial [\mathbf{W}_{n-1}\sigma(\mathbf{W}_{n-2}\vec{a}_{n-2})]}{\partial \mathbf{W}_{n-2}} \\ &= 2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e}) \odot \sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1})\mathbf{W}_{n-1}\sigma'(\mathbf{W}_{n-2}\vec{a}_{n-2})\vec{a}_{n-2} \end{aligned}$$

This is quite messy, but the chain rule is done properly. Then, correct for dimensions:

$$\begin{aligned} \frac{\partial C}{\partial \mathbf{W}_{n-2}} &= \underbrace{2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e})}_{q\times 1} \odot \underbrace{\sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1})}_{q\times 1} \underbrace{\mathbf{W}_{n-1}}_{q\times p} \underbrace{\sigma'(\mathbf{W}_{n-2}\vec{a}_{n-2})}_{p\times 1} \underbrace{\vec{a}_{n-2}}_{r\times 1} \\ &= \left[\underbrace{\mathbf{W}_{n-1}}_{p\times q}^{\mathrm{T}} \underbrace{2(\sigma(\mathbf{W}_{n-1}\vec{a}_{n-1}) - \vec{e})}_{q\times 1} \odot \underbrace{\sigma'(\mathbf{W}_{n-1}\vec{a}_{n-1})}_{q\times 1} \right] \odot \underbrace{\sigma'(\mathbf{W}_{n-2}\vec{a}_{n-2})}_{p\times 1} \underbrace{\vec{a}_{n-2}}_{1\times r}^{\mathrm{T}} \\ \Rightarrow \underbrace{\frac{\partial C}{\partial \mathbf{W}_{n-2}}}_{p\times r} &= \mathbf{W}_{n-1}^{\mathrm{T}}\delta_{n-1} \odot \sigma'(\mathbf{W}_{n-2}\vec{a}_{n-2})\vec{a}_{n-2}^{\mathrm{T}} \\ &= \delta_{n-2}\vec{a}_{n-2}^{\mathrm{T}} \end{aligned}$$

Or, in general:

$$\frac{\partial C}{\partial \mathbf{W}_j} = \delta_j \vec{a}_j^{\mathrm{T}},$$

where,

$$\delta_i = \begin{cases} 2(\sigma(\mathbf{W}_i\vec{a}_i) - \vec{e}) \odot \sigma'(\mathbf{W}_i\vec{a}_i) & i = n-1 \\ \mathbf{W}_{i+1}^{\mathrm{T}}\delta_{i+1} \odot \sigma'(\mathbf{W}_i\vec{a}_i) & i < n-1. \end{cases}$$

Heavily inspired by https://sudeepraja.github.io/Neural/